

Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Package A1111 | Release 5

AUTOMATED PREPARATION CHECKS
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

DOCUMENT BOUNDARY

This package supports preparation for independent verification. It is not a verification opinion, accreditation decision, customs decision, CBAM Registry submission or acceptance guarantee.

Control field			
Case ID	case_verifier_grade_fixture	Generated	2027-01-15T12:00:00.000Z
Ruleset	3.0.0	Source registry	CBAM-EU-2026.07.16
Root hash	179d2547250f42d58eb1b1a3c79283294ad74c4f4568871a6ff6fed5d94f5418	Verifier status	NOT_REVIEWED

Product Scope - Controlled identity

Field	
Importer	CBAMValid Importer B.V.
EORI	NL123456789AB
Exporter / operator	Verified Steel Operator A.S.
Installation	Verified Integrated Steel Installation
Country	TR
Production route	Blast Furnace Route (BF-BOF)
Reporting period	2026-ANNUAL

Product Scope - Goods population

Good							
1	72011011	IRON_AND_STEEL	60	0.6	48	0.8	0.04
2	72011019	IRON_AND_STEEL	40	0.4	32	0.8	0.04

Product Scope - Scope conclusion

AUTOMATED READINESS
READY_FOR_INDEPENDENT_VERIFICATION

Product Scope - Legal basis

Source			
REG_2023_956	32023R0956	2023-05-17	CBAM framework; Annex I goods scope; Annex IV embedded-emissions calculation framework

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Source			
REG_2025_2083	32025R2083	2025-10-20	50 tonne annual mass threshold for cement, fertilisers, iron and steel, and aluminium; definitive-period declaration and certificate obligations; actual values require independent verification
IMPL_2025_2546	32025R2546	2026-01-01	reasonable assurance and risk-based verification; 5 percent per-good materiality levels; electronic verification report template
IMPL_2025_2547	32025R2547	2026-01-01	functional units and production processes; monitoring plan minimum elements; sector-specific system boundaries and embedded-emissions methods
IMPL_2025_2548	32025R2548	2026-01-01	calculation and publication of CBAM certificate prices
DEL_2025_2551	32025R2551	2026-01-01	accreditation and competence of verifiers; verification planning, evidence and site-visit framework; oversight and independence

CN Code - Classification register

Good				
1	72011011	IRON_AND_STEEL	11111111-1111-4111-8111-222222222222	11111111-1111-4111-8111-444444444444
2	72011019	IRON_AND_STEEL	11111111-1111-4111-8111-222222222222	11111111-1111-4111-8111-444444444444

CN Code - Classification boundary

This document records the classification supplied in the sealed case. It does not replace a binding customs classification decision. Any disputed or uncertain CN code remains subject to customs review.

Checklist - Monitoring-plan requirements

Requirement			
MP-01	Monitoring-plan date, version and responsible operator identity	DOCUMENTE D	Case version 1; owner recorded
MP-02	Installation description and production processes	DOCUMENTE D	Blast Furnace Route (BF-BOF)
MP-03	CN-coded goods and functional production units	DOCUMENTE D	2 good(s) recorded
MP-04	System boundaries and directly connected processes	DOCUMENTE D	Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.
MP-05	Source streams, emission sources and measurement methods	DOCUMENTE D	0 precursor(s); direct and electricity inputs recorded
MP-06	Data-source hierarchy, corroboration and control activities	DOCUMENTE D	4 approved and clean evidence file(s)
MP-07	Calculation methods, conversions and rounding	DOCUMENTE D	10 hashed trace node(s)
MP-08	Goods allocation method and reconciliation	DOCUMENTE	Allocation delta 0

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Require ment			
		D	
MP-09	Precursor scope and embedded emissions treatment	DOCUMENTE D	Accepted no-precursor decision
MP-10	Methodology deviations, estimates and technical justifications	DOCUMENTE D	2/2 methodology decision(s) accepted
MP-11	Per-good materiality assessment	DOCUMENTE D	5% per-good specific-emissions threshold
MP-12	Evidence integrity and immutable calculation provenance	DOCUMENTE D	Calculation root 179d2547250f42d58eb1b1a3c79283294ad74c4f4568871a6ff6fed5d94f5418

Checklist - Automated control findings

Rule				
QC_SCENARIO	Illustrative scenario replacement	NOT_APPLICABLE	No automated finding	—
QC_00	Operator, installation and boundary identity	PASS	No automated finding	REM_COMPLETE_CASE_IDENTITY
QC_00_ORIGIN	Installation country of origin CBAM scope	PASS	No automated finding	—
QC_01	EORI format and evidence	PASS	No automated finding	—
QC_02	Definitive-period reporting year	PASS	No automated finding	REM_CORRECT_REPORTING_YEAR
QC_03_0	Good 1 CN code and evidence	PASS	No automated finding	—
QC_04_0	Good 1 production and evidence	PASS	No automated finding	—
QC_05_0	Good 1 sector	PASS	No automated finding	REM_SELECT_SUPPORTED_SECTOR
QC_03_1	Good 2 CN code and evidence	PASS	No automated finding	—
QC_04_1	Good 2 production and evidence	PASS	No automated finding	—
QC_05_1	Good 2 sector	PASS	No automated finding	REM_SELECT_SUPPORTED_SECTOR
QC_05A	Allocation, evidence and reconciliation	PASS	No automated finding	—
QC_06	directEmissions value, unit and evidence	PASS	No automated finding	—
QC_07	electricityConsumed value, unit and evidence	PASS	No automated finding	—
QC_08	gridEmissionFactor value, unit and evidence	PASS	No automated finding	—
QC_09	Precursor scope	PASS	No automated finding	REM_CONFIRM_PRECURSOR_SCOPE
QC_10	Evidence integrity	PASS	No automated finding	—
QC_11	Carbon price records	NOT_APPLICABLE	No automated finding	—

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Rule				
		BLE		
QC_12	Goods cross-artifact consistency	PASS	No automated finding	—

Checklist - Evidence coverage

Measure	
Registered evidence files	4
Approved and malware-clean files	4
Linked input fields	18
Linked calculation nodes	5
Coverage rate	100%
Duplicate hashes	0

Monitoring Plan - Installation and boundary

Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

Production route: Blast Furnace Route (BF-BOF)

Monitoring Plan - Minimum monitoring-plan elements

Require ment			
MP-01	Monitoring-plan date, version and responsible operator identity	DOCUMENTE D	Case version 1; owner recorded
MP-02	Installation description and production processes	DOCUMENTE D	Blast Furnace Route (BF-BOF)
MP-03	CN-coded goods and functional production units	DOCUMENTE D	2 good(s) recorded
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MP-05	Source streams, emission sources and measurement methods	DOCUMENTE D	0 precursor(s); direct and electricity inputs recorded
MP-06	Data-source hierarchy, corroboration and control activities	DOCUMENTE D	4 approved and clean evidence file(s)
MP-07	Calculation methods, conversions and rounding	DOCUMENTE D	10 hashed trace node(s)
MP-08	Goods allocation method and reconciliation	DOCUMENTE D	Allocation delta 0
MP-09	Precursor scope and embedded emissions treatment	DOCUMENTE D	Accepted no-precursor decision

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Require ment			
MP-10	Methodology deviations, estimates and technical justifications	DOCUMENTE D	2/2 methodology decision(s) accepted
MP-11	Per-good materiality assessment	DOCUMENTE D	5% per-good specific-emissions threshold
MP-12	Evidence integrity and immutable calculation provenance	DOCUMENTE D	Calculation root 179d2547250f42d58eb1b1a3c79283294ad74c4f4568871a6ff6fed5d94f5418

Monitoring Plan - Input hierarchy

Input					
Direct emissions	80	tCO2e	PRIMARY	11111111-1111-4111-8111-5555555555	Documented direct measurement and reconciled production ledger
Electricity	100	MWh	PRIMARY	11111111-1111-4111-8111-3333333333	Documented direct measurement and reconciled production ledger
Grid emission factor	0.4	tCO2e/MWh	PRIMARY	11111111-1111-4111-8111-3333333333	Documented direct measurement and reconciled production ledger

Process Map - End-to-end process

Stage		
1	Identify installation, operator, CN-coded goods and reporting period	CASE and GOODS registers
2	Define production processes, source streams, boundaries and functional units	Monitoring Plan and System Boundary
3	Capture direct, electricity and precursor data with approved evidence	INPUTS, PRECURSORS and EVIDENCE
4	Apply unit conversions, allocation and embedded-emissions formulae	CALCULATION_TRACE and root hash
5	Run evidence, methodology, reconciliation and integrity controls	QUALITY_CONTROLS and MONITORING_PLAN
6	Apply per-good 5% materiality reference for verifier planning	GOODS materiality columns
7	Freeze, hash, sign and commit immutable package	Manifest, KMS signature and package hash
8	Independent accredited verifier performs review and records opinion	VERIFIER_SIGN_OFF — initially NOT_REVIEWED

System Boundary - Declared system boundary

BOUNDARY

Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

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System Boundary - Sector methodology

Sector			
Iron and Steel	IN_FORCE	Coke oven, sinter or pellet preparation, blast furnace, basic oxygen furnace, direct reduction, electric arc furnace, casting and finishing processes within the installation boundary.	Mass balance and production ledger; Fuel and reducing-agent consumption; Precursor embedded emissions; Waste-gas and transferred-heat treatment; Allocation to CN-coded goods

Methodology - Decision register

Topic					
PRECURSOR_SC OPE	No qualifying precursor line is declared for the fixture goods and controlled route.	The controlled bill of materials and production route contain no separate precursor goods requiring an additional precursor line in this fixture.	Regulation (EU) 2023/956 Annex IV and Commission Implementing Regulation (EU) 2025/2547; supported by the controlled bill of materials.	ACCEPTED	11111111-1111-4111-8111-555555555555
GOODS_EMISSIONS_ALLOCATION	Allocate installation emissions using documented product mass shares of 0.6 and 0.4.	The two goods are produced in the same controlled period and the reconciled product ledger provides complete mass shares summing to one.	Commission Implementing Regulation (EU) 2025/2547 allocation and monitoring-plan requirements; controlled production ledger.	ACCEPTED	11111111-1111-4111-8111-444444444444

Calculation Annex - Engine identity

Control	
Ruleset	EU-CBAM-DEFINITIVE-2026
Engine version	3.0.0
Calculation root hash	179d2547250f42d58eb1b1a3c79283294ad74c4f4568871a6ff6fed5d94f5418
Allocation share total	1
Allocation reconciliation delta	0

Calculation Annex - Formula trace

Formula				
CBAM_INSTALLATION_DIRECT_EMISSIONS	80	tCO2e	690085cd04f6102b1d2617fa92b671e089632e2db84f8c657a521938be42eb57	None
CBAM_INDIRECT_EMISSIONS	40	tCO2e	98a76aac49f58cbd4c0ad68637e03e592b2dd2e0bc7208a8062648de29ffe081	None
CBAM_PRECURSOR_EMISSIONS_SUM	0	tCO2e	5553fb3ff7a2d2d3f04af35bdc6f4c64f77ee3359321812bc30163f210d3c86	None
CBAM_TOTAL_EMBEDDED_EMISSIONS_DISCLOSED	120	tCO2e	3bd0d06237ae6b808a635fb731efb40b74fb90a51e42bfe49ed064cb41f03e2f	Disclosed total combines direct and indirect emissions before

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Formula				
				sector-specific pricing rules.
CBAM_GOOD_EMISSIONS_ALL OCATION_1	0.8	tCO2e/t	137fc83c1e53a213bd38caccb435960763805 0549305b90e340da30d856a26c1	None
CBAM_GOOD_EMISSIONS_ALL OCATION_2	0.8	tCO2e/t	2670d4a69b964809495520ced63320b6cced 57f17d354d9a422aaf1b7c7df0e9	None
CBAM_TOTAL_EMBEDDED_EMI SSIONS_PRICED	80	tCO2e	1b547bb68197cb8f0357a93b9cd6f8655d4ca c8c062d1840050938f4a7d82966	Priced total applies sector Annex II direct-only rules where applicable.
CBAM_GOODS_ALLOCATION_R ECONCILIATION	0	tCO2e	a507c7fe68ea6bff1451969dc6b94557b31668 9ea670e3948498cd1d3750091b	None
CBAM_AGGREGATE_SPECIFIC_ EMBEDDED_EMISSIONS	0.8	tCO2e/t	ead9fddcade625553ba96d6a41e1913ed8e8a d1c03b6cf1b955e59c5272b795e	The aggregate intensity is a portfolio diagnostic; reportable values are the per-good intensities.
CBAM CARBON PRICE REDU CTION_RECORD_TOTAL	0	certificate-equ ivalent	13eee823710e59782162be2fd529ca7dd9bb3 ac7b195d5d21486017266c848ea	None

Calculation Annex - Per-good reconciliation

Good							
1	72011011	IRON_AND_STEEL	60	0.6	48	0.8	0.04
2	72011019	IRON_AND_STEEL	40	0.4	32	0.8	0.04